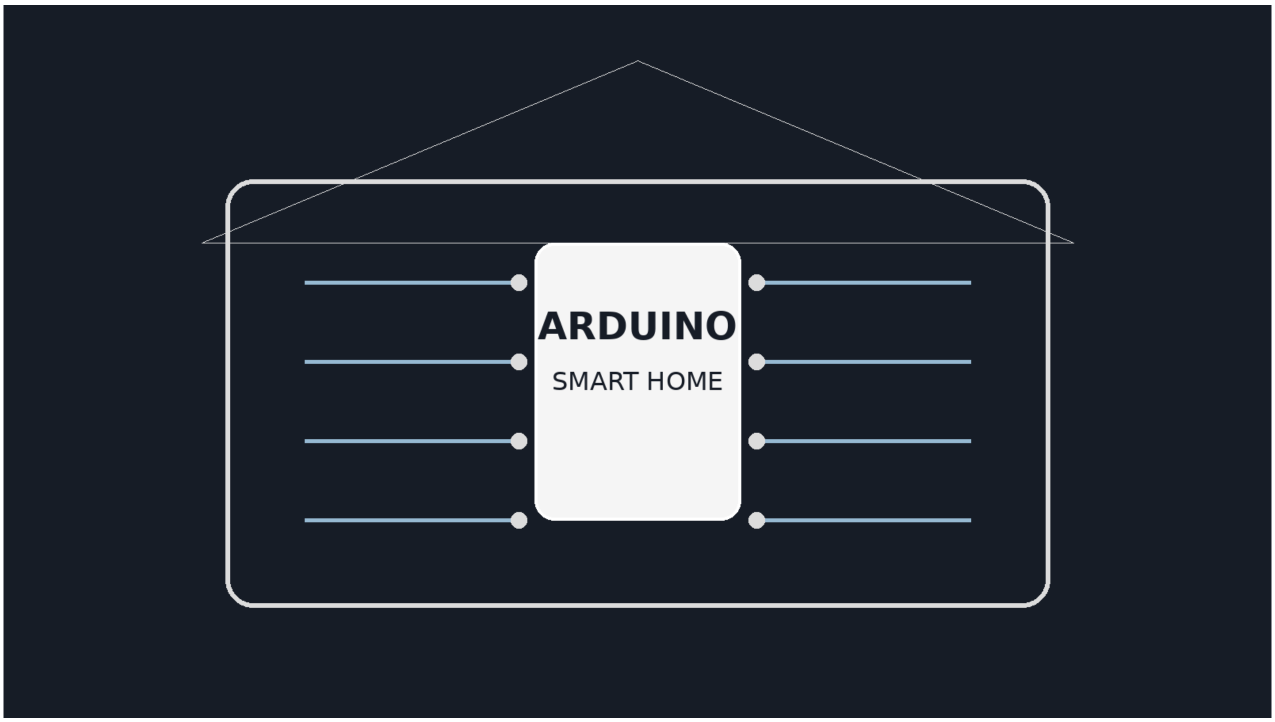
SMART HOME AUTOMATION SYSTEM

Using Arduino



PROJECT REPORT

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Project	Arduino-Based Smart Home Automation

Abstract

This report presents the design of a basic Smart Home Automation System using an Arduino Uno. The proposed system uses sensors to observe conditions such as motion, ambient light, gas concentration, or temperature. Arduino processes these inputs and controls appliances through a relay module. The prototype demonstrates how low-cost embedded hardware can improve comfort, safety, and energy efficiency. The design follows a modular approach, allowing future expansion through Bluetooth, Wi-Fi, mobile applications, cloud dashboards, voice commands, and intelligent automation.

Keywords: *Arduino Uno, smart home, automation, sensors, relay, Internet of Things, energy efficiency.*

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1. Introduction

Home automation refers to the automatic or remote control of household devices. In this project, Arduino Uno acts as the central controller. Sensors provide information from the environment, while a relay module acts as an electrical switch for connected loads. The Arduino Uno is suitable for educational prototypes because it provides digital and analog input/output pins, USB programming, and a simple development environment.

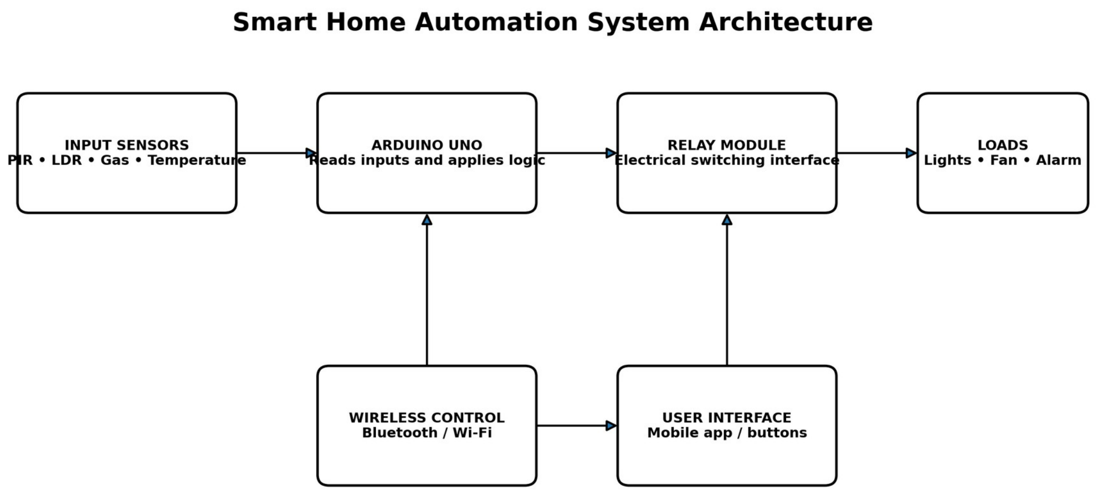

Figure 1. Overall architecture of the proposed smart home system.

2. Problem Statement

Conventional homes depend on manual switching and continuous human attention. Appliances may remain active when rooms are empty, creating unnecessary energy use. Safety conditions such as gas leakage, overheating, or unexpected movement may also remain undetected. A low-cost automation system is therefore needed to monitor selected conditions and control devices automatically.

Problem	Effect	Proposed Response
Manual switching	Reduced convenience	Automatic sensor-based control
Devices left ON	Energy wastage	Time/condition-based switching
Slow hazard detection	Safety risk	Gas, motion or temperature alerts
No remote access	Limited control	Bluetooth/Wi-Fi extension

3. Project Objectives

- Design a basic Arduino-based automation prototype.
- Control at least one electrical load through a relay module.
- Read input from motion, light, gas, or temperature sensors.
- Apply programmed conditions to decide when a device should switch ON or OFF.
- Improve convenience, safety, and energy awareness.
- Keep the design modular so wireless and IoT features can be added later.

4. Scope and Requirements

The project is intended as an educational low-voltage prototype. It demonstrates control logic and system integration rather than a complete commercial installation. The initial prototype can use a PIR sensor for motion detection, an LDR for light detection, a relay module, and a low-voltage lamp or LED as the controlled load.

Functional Requirements	Non-Functional Requirements
Read sensor inputs	Low cost and easy to assemble
Execute decision logic	Safe low-voltage demonstration
Switch a load using a relay	Reliable and understandable operation
Provide status through LEDs or Serial Monitor	Expandable for future features

5. Hardware and Software Components

Component	Purpose	Typical Interface
Arduino Uno	Central microcontroller board that reads inputs and produces control outputs.	USB / digital / analog
PIR Motion Sensor	Detects movement in a room or monitored area.	Digital input
LDR and Resistor	Forms a voltage divider to measure ambient light level.	Analog input
Gas or Temperature Sensor (Optional)	Adds environmental monitoring and warning capability.	Analog or digital input
Relay Module	Provides an isolated switching interface between Arduino logic and a load.	Digital output
Demo Lamp / LED / Fan	Represents the appliance controlled by the system.	Relay contact
Breadboard and Jumper Wires	Used to create the prototype connections.	Physical wiring
5 V Power Supply / USB Cable	Powers the Arduino and low-voltage modules.	5 V supply
Arduino IDE	Used to write, compile, upload, and monitor the program.	Computer software

The Arduino Uno R3 is based on the ATmega328P and includes 14 digital input/output pins and 6 analog inputs, which is sufficient for a small multi-sensor prototype.

6. Working Principle

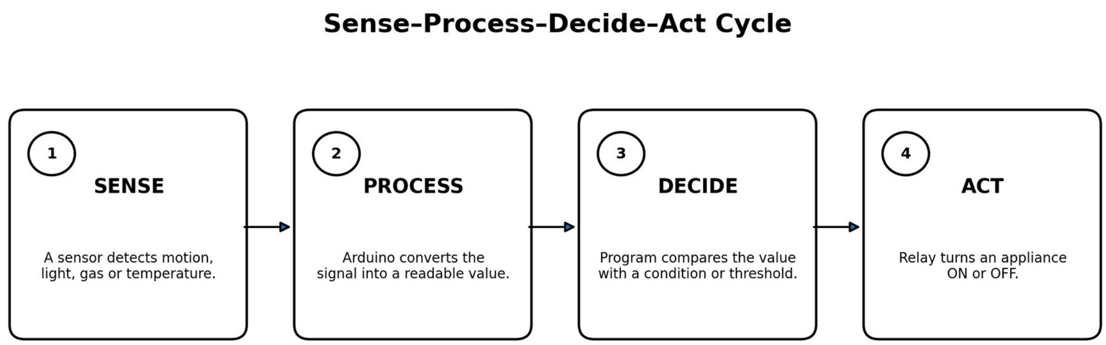

Figure 2. Basic operating cycle of the automation system.

The system continuously reads one or more sensors. The program compares each reading with a predefined rule. For example, a light may be switched ON only when motion is detected and the ambient light is below a selected threshold. When the condition is no longer true, the system can switch the light OFF after a short delay.

7. Prototype Design and Implementation

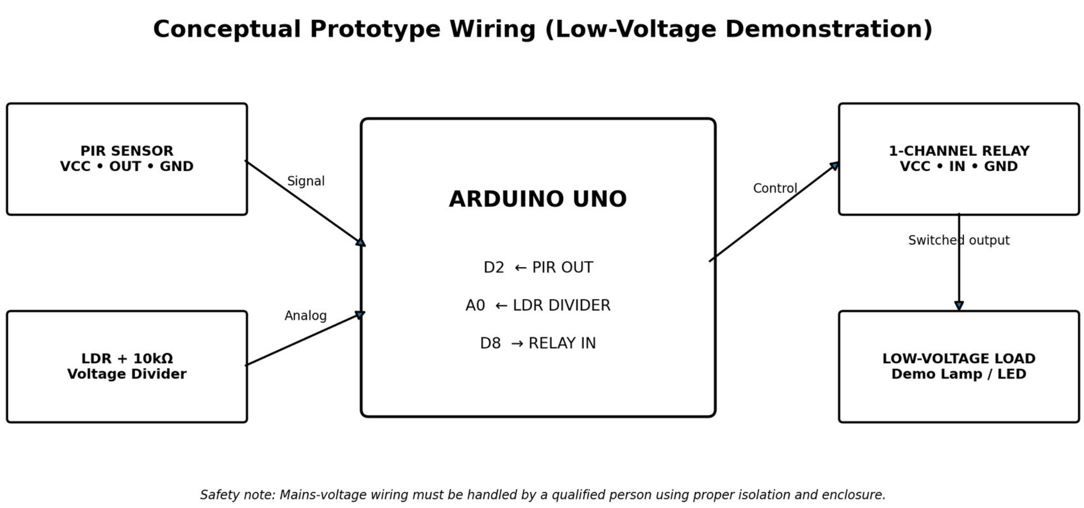

Figure 3. Conceptual low-voltage wiring for the prototype.

1. Connect the PIR sensor output to a digital input pin.
2. Create an LDR voltage divider and connect its midpoint to an analog input.
3. Connect the relay control input to an Arduino digital output.
4. Use a separate, suitable power arrangement for the demonstration load.
5. Upload the Arduino sketch and open the Serial Monitor for testing.
6. Adjust the light threshold and delay according to the room conditions.

8. Sample Arduino Program

The following illustrative code demonstrates automatic light control using a PIR sensor, an LDR, and a relay. Pin numbers and relay logic may need adjustment for the selected hardware.

```
const int PIR_PIN = 2;
const int LDR_PIN = A0;
const int RELAY_PIN = 8;

const int DARK_THRESHOLD = 450;
unsigned long lastMotionTime = 0;
const unsigned long OFF_DELAY_MS = 15000;

void setup() {
  pinMode(PIR_PIN, INPUT);
  pinMode(RELAY_PIN, OUTPUT);
  digitalWrite(RELAY_PIN, LOW);
  Serial.begin(9600);
}

void loop() {
  int motion = digitalRead(PIR_PIN);
  int lightLevel = analogRead(LDR_PIN);

  if (motion == HIGH) {
    lastMotionTime = millis();
  }

  bool roomIsDark = lightLevel < DARK_THRESHOLD;
  bool recentMotion = (millis() - lastMotionTime) < OFF_DELAY_MS;

  digitalWrite(RELAY_PIN, (roomIsDark && recentMotion) ? HIGH : LOW);

  Serial.print("Motion: ");
  Serial.print(motion);
  Serial.print(" | Light: ");
  Serial.println(lightLevel);

  delay(100);
}
```

9. Testing Plan

Test	Method	Expected Result
PIR detection	Move in front of the sensor	Motion value changes and timer is updated
Light sensing	Cover and uncover the LDR	Analog reading changes significantly
Automatic ON	Create darkness and motion	Relay/load turns ON
Automatic OFF	Stop movement and wait for delay	Relay/load turns OFF
False trigger check	Observe without movement	System remains stable
Power restart	Disconnect and reconnect power	System starts in a safe state

10. Applications

Residential rooms: Automatic lighting, fan control, door monitoring and safety alerts.

Offices: Occupancy-based lighting and energy-saving controls.

Classrooms: Automatic lights, projector support and environmental monitoring.

Elderly assistance: Simple controls, alarms and activity detection.

Security systems: Motion-based alarms and remote notifications.

Energy monitoring: Collection of operating data for better usage decisions.

11. Advantages, Limitations and Safety

Advantages	Limitations	Safety Considerations
Affordable prototype	Sensor accuracy varies	Use low-voltage loads during testing
Easy to program	Basic logic is not intelligent	Do not expose live terminals
Reduces manual effort	Wireless control requires extra modules	Use relay isolation and enclosure
Modular and expandable	Power failure stops operation	Separate logic and load wiring
Supports energy saving	Prototype is not a certified product	Qualified personnel for mains wiring

Important: This academic prototype should be demonstrated with a low-voltage lamp or LED. Direct connection to household mains electricity can cause electric shock, fire, or equipment damage and must not be attempted without proper protection and qualified supervision.

12. Future Scope

- Wi-Fi or Bluetooth control through a smartphone.
- Arduino Cloud or another IoT dashboard for remote monitoring.
- Voice control through supported assistant platforms.
- Door locks, curtains, irrigation, smoke alarms, and appliance scheduling.
- Energy metering and daily consumption reports.
- Machine-learning-based routines that adapt to user behavior.
- Battery backup and fault notifications for improved reliability.

13. Conclusion

The Smart Home Automation System using Arduino demonstrates a practical method for combining sensors, programmed logic, and relay-based switching. The system can automatically respond to room conditions, reduce unnecessary manual operation, and support energy-saving and safety functions. Because the design is modular, the same basic prototype can be expanded into a more advanced IoT-based solution. The project also provides valuable experience in embedded programming, sensor interfacing, control logic, testing, and system integration.

14. References

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Appendix A: Suggested Bill of Materials

Item	Quantity
Arduino Uno compatible board	1
PIR motion sensor	1
LDR photoresistor	1
10 kΩ resistor	1
1-channel 5 V relay module	1
Low-voltage LED/lamp	1
Breadboard	1
Jumper wire set	1 set
USB cable / 5 V supply	1